UNIVERSITY OF WOLLONGONG AUSTRALIA

Project 1

Vehicle Seat Suspension

Individual project

100 marks

MATLAB submission: plots and modified scripts for **Stages 1, 4 and 5**.

A utility vehicle crosses a raised joint in a test road (**Figure 1**). The cab floor drives the seat through springs and a damper. Begin with the seat design shown below, then investigate changes to its damping in the later tasks. (Use content from Lectures 1–5; do not model vehicle pitch, wheels or driver flexibility.)

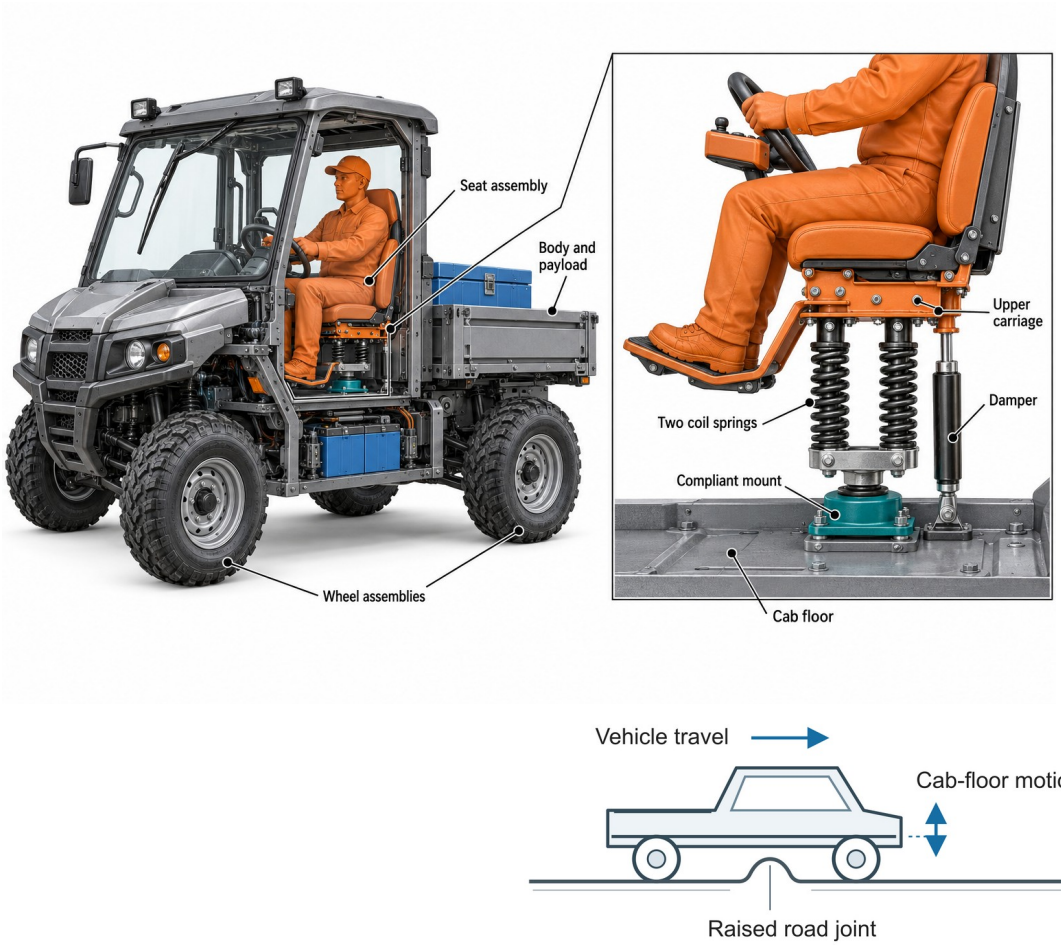


Figure 1. Vehicle and seat context. The small sketch shows a raised road joint; the model input is cab-floor motion. Orange components move with the seat; the lower pedestal follows the floor.

Given model and test

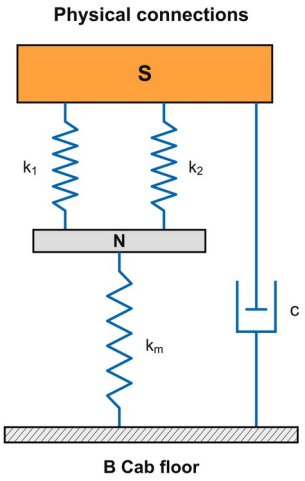


Figure 2. Physical nodes: S is the moving seat, N the massless plate, and B the cab floor. The damper connects node S directly to node B.

Here x(t) is the vertical displacement of the moving seat (node S), and r(t) is the prescribed vertical displacement of the cab floor (node B). Both are measured from loaded static equilibrium. Suspension travel is the displacement of the seat relative to the cab floor. $z = x - r$. For the floor-bump tests, initially $x = \dot{x} = r = \dot{r} = 0$. Springs and damper are linear; the driver moves rigidly with the seat. Take upward x and r as positive.

Table 1. Spring and damper connections and values.

Element	Connections (nodes)	Value
Coil 1 and coil 2	Both S to N	12 kN/m each
Mount	N to B	72 kN/m
Damper	S directly to B	1.5 kN·s/m

Parameter values are given in kN-based units. Convert to N-based SI units for Newton's law and MATLAB.

Table 2. Test conditions and design limits.

Test quantity	Condition or requirement
Floor input	One smooth bump, peak 30 mm, duration 0.35 s; zero before and after.
Floor-bump window	0–3 s, including the response after the bump.
Dynamic acceleration	Limit the acceleration felt by the driver. Peak magnitude ≤ 0.53g, excluding gravity; g = 9.81 m/s².
Relative travel	Prevent the suspension from hitting its end stops. Peak $\|z\| \le 45$ mm.

In the input formula below, r is in metres and t is in seconds.

$$r(t) = 0.015\left[1 - \cos(\alpha t)\right],$$

$$\alpha = \frac{2\pi}{0.35}\ rad/s, \qquad 0 \le t \le 0.35\ s$$

$$r = \dot{r} = 0 \ \ outside\ \ the\ \ bump$$

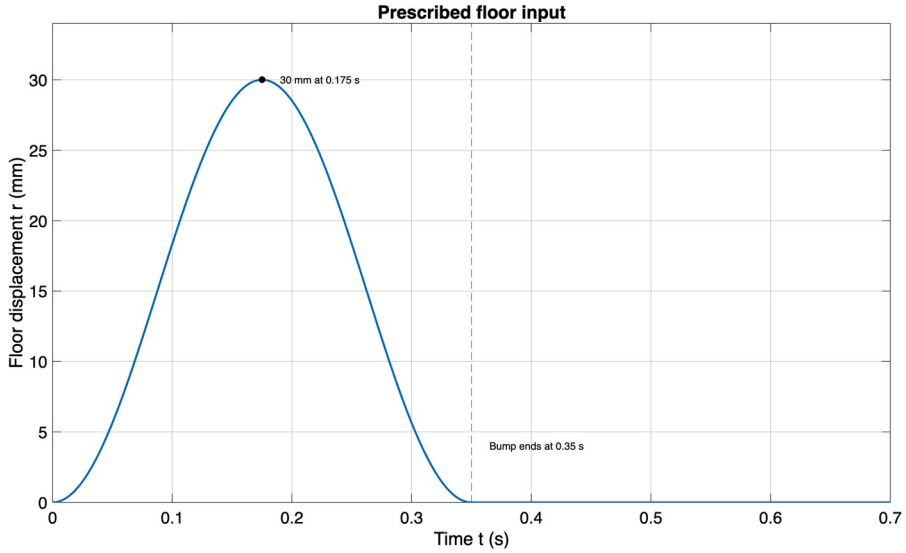


Figure 3. MATLAB plot of the prescribed floor displacement. Peak 30 mm at 0.175 s; duration 0.35 s. Source: MATLAB/Project01_01_Input.m.

Stage 1 Physical model after Lecture 1 (20 marks)

Table 3. Component masses and attachment locations.

Component	Mass kg	Attached to
Driver including clothing	68	Moving seat
Seat shell cushion and backrest	7	Moving seat
Upper carriage and moving rails	3	Moving seat
Armrest controls and footrest	2	Moving seat
Chassis cab and drive units	520	Vehicle body
Battery pack	110	Vehicle body
Tools in secured box	32	Vehicle body
Other fixed equipment	46	Vehicle body
Lower pedestal	8	Vehicle body
Fixed seat hardware	4	Vehicle body
Four wheel/hub assemblies	25 each	Wheels

Use **Figure 2**, **Table 1** and **Table 3**.

1. Calculate the equivalent moving mass m.
2. Determine the equivalent spring stiffness k and damping coefficient c.
3. Draw the single-mass equivalent seat model. Label m, k, c, x, r and the positive direction.
4. Draw the free-body diagram of the moving mass. Show the spring and damper forces.

5. Name the prescribed input and the two independent states. State the model order and define relative travel z.

6. Modify MATLAB/Project01_01_Input.m to plot floor displacement (mm) and floor velocity (mm/s) over 0–0.7 s in two labelled panels. Use its r and rd arrays. Submit the plot and your modified script.

Hint: use `subplot(2,1,1)` for the upper panel and `subplot(2,1,2)` for the lower panel.

7. From your plots, find the peak displacement, maximum and minimum velocity, and their corresponding times. Present these values in a table. Check them using the supplied bump formula and verify that both signals are zero at the bump end.

Stage 2 Equation and Laplace transform after Lecture 2 (15 marks)

Design A is the original seat specified in **Tables 1 and 3**, with c = 1.5 kN·s/m. During service, its hydraulic damper develops an oil leak. For this fault test, assume it loses all damping force, so c = 0. This failed-damper configuration is Design B; its mass and springs are unchanged.

Use the equivalent model and free-body diagram from **Stage 1, Questions 1–4**. Coordinates remain measured from loaded static equilibrium, so gravity cancels from the incremental force balance.

1. From your free-body diagram, write Newton's law for Design A. Express spring and damper forces using relative displacement and velocity, collect the x terms, and give the numerical ODE.

2. Then write the modified ODE for Design B without repeating the full derivation.

3. Take the Laplace transform of the ODE for Design A and collect X(s). Use the stated zero initial conditions.

Stage 3 Transfer function after Lecture 3 (20 marks)

1. From the Stage 2 force balance, draw a mechanical feedback block diagram for Design A, with floor displacement r as input and seat displacement x as output. Label the signals, block transfer functions and summing signs.

2. Simplify your block diagram to obtain the transfer function $H_x(s) = X(s)/R(s)$. Give the Design A coefficients.

3. Calculate $H_x(0)$, and explain the slow-motion result.

Stage 4 Parameters and response after Lecture 4 (35 marks)

Stage 4.1: Fixed-floor release test

A maintenance technician proposes two replacement dampers: Design C and Design D, specified in **Table 4**. Both are fitted to the original seat of Design A, with its mass and springs unchanged. Compare them with Design A and the failed-damper Design B in the release test. Use Designs A–C for the later floor-bump and frequency tests.

For this release test, fix the floor at zero. Use the mass and stiffness from Stage 1. Compare the four designs in **Table 4**. Start each seat 10 mm above equilibrium and release it from rest. This is a zero-input free response caused by the initial displacement.

1. Compare the characteristic polynomial with the standard second-order form to determine the common natural frequency ω_n and the damping ratio ζ of each design. Complete **Table 4** and classify the cases as undamped, underdamped, critically damped or overdamped.

2. For Design A only, calculate its two poles and damped oscillation period T_d. Plot the simulated displacement over 0–1.5 s and use this time-domain response to measure the oscillation period, zoom in if required. Compare the calculated and measured periods. (*Reference: Lecture 4, slide 11 Pole location and damping regime).*

3. For Design B, use the same fixed-floor release conditions. Apply the Laplace transform with the initial displacement retained and find x(t). State its amplitude and period, then check them against the Design B curve in Question 4.

4. Modify MATLAB/Project01_04_FreeDecay.m to include Designs A–D with their damping values from **Table 4**. Plot all four displacement responses in a single figure over 0–1.5 s, with displacement in mm and a legend identifying each design. Submit the figure and your modified script.

Table 4. Original, failed-damper and replacement designs; mass and springs unchanged.

Design	c (kN·s/m)	ω_n (rad/s)	ζ	Type
Design A	1.5			
Design B	0			
Design C	2.4			
Design D	3.6			

Stage 4.2: Floor-bump test

Figure 4 and **Table 5** show the supplied results for Designs A and B. Add Design C using its parameters in **Table 4**, then compare all three designs under the prescribed floor bump. Use zero initial displacement and velocity.

Table 5. Supplied response peaks for Design A and Design B.

Design	Peak |a|/g	Peak |z| mm
Design A	0.54	14.0
Design B	1.13	49.1

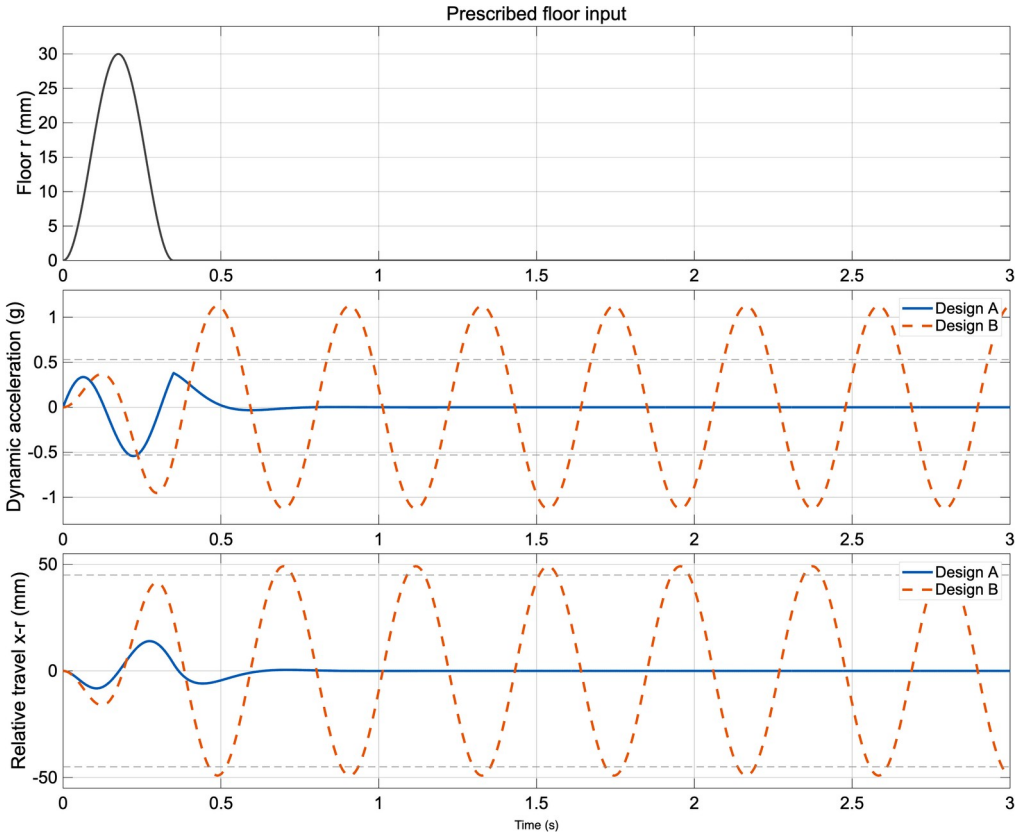


Figure 4. MATLAB ode45 responses for Design A and Design B. Dynamic acceleration is in g; relative travel is x − r in mm. Dashed lines show the two limits. Source: MATLAB/Project01_02_Response.m.

5. Add Design C to MATLAB/Project01_02_Response.m using **Table 4**. Update the case labels and legends. Plot the A–C dynamic acceleration (g) and relative travel (mm) in two panels of one figure over 0–3 s, with both limits shown. Submit the figure and your modified script.

6. Submit a table of peak acceleration (g) and peak relative travel (mm) for Designs A–C. Identify which design has the smallest peak acceleration and which has the smallest peak travel.

Stage 5 Frequency reading and decision after Lecture 5 (10 marks)

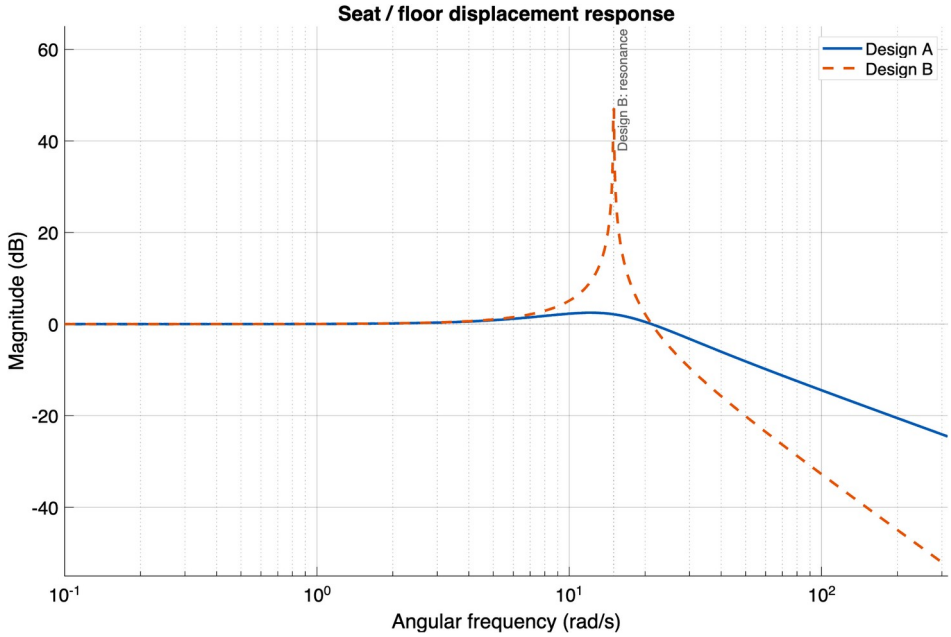


Figure 5. Bode magnitude plots for Designs A and B. Frequency uses a logarithmic scale; magnitude in dB uses a linear scale. Design B has resonance at 15 rad/s. MATLAB: Project01_03_Frequency.m.

Table 6. Supplied displacement gains at 60 rad/s (dB).

Frequency	Design A (dB)	Design B (dB)
60 rad/s	−9.83	−23.52

1. Add Design C to MATLAB/Project01_03_Frequency.m. Submit one Bode magnitude plot comparing Designs A–C over 0.1 to $10^{2.5}$ rad/s, and a table of their gains at 60 rad/s in dB. Use a logarithmic frequency axis and a linear dB axis. Include your modified script.

2. State the low-frequency limit. Compare the resonant behaviour of Designs A–C, then rank their transmission at 60 rad/s using your table. Explain the damping trade-off shown by these comparisons.

For undamped Design B, the vertical line at 15 rad/s marks resonance; the sampled curve does not give a finite resonant maximum.

3. Keep Design C and zero initial conditions. Increase only the bump height by 5%, from 30 to 31.5 mm, keeping its shape and 0.35 s duration. Before calculating or simulating, predict the changes in peak acceleration, peak relative travel and the seat-to-floor transfer function. Explain your reasoning.

4. Complete **Table 7** for Designs A–C using your unrounded peaks from Stage 4, Questions 5–6 and the limits in **Table 2**. Use the original 30 mm bump for this table.

5. Calculate Design C's acceleration margin below 0.53g and travel clearance below 45 mm for the original bump. Quantify both peaks for the 5% higher bump and check the same limits. Recommend whether C is acceptable for each test. State one physical check or measurement needed before recommending a real seat. No additional plot is required.

Hint. Compare each criterion separately: acceleration ≤ 0.53g and travel ≤ 45 mm. A lowest ranking is not a pass.

Table 7. Acceleration and travel checks for Designs A–C.

Design	Acceleration pass/fail	Travel pass/fail	Overall
Design A			
Design B			
Design C			

Submission and marking

Submit one individual report by Stage and question, showing calculations and completed tables.

Report acceleration to 2 decimal places in g, and displacement and velocity to 1 decimal place in mm and mm/s. Use up to 3 decimal places for short peak times, 3 significant figures for calculated angular frequencies and damping ratios, and 2 decimal places for frequency-response gains in dB. Keep full precision in calculations; round only reported results.

Include the new plots and modified MATLAB files for **Stages 1, 4 and 5**. Label axes with units and use legends.

Identify supplied material and any software or AI used, and briefly state how you checked its output. Where course rules permit, these tools may assist algebra, coding and plotting; you remain responsible for the model, physical reasoning and verification. Full AI conversation histories are not required.

Due: the published Project 1 cutoff, after Lecture 5 content has been completed. Total: 100 marks.

Table 8. Stage marks (100 marks total).

Stage	Marks
Stage 1	20
Stage 2	15
Stage 3	20
Stage 4	35
Stage 5	10
Total required marks	**100**

Method and sources

Adapted from current MECH321 Lectures and Tutorials. Parameters and limits are teaching data; curves are model predictions.

Prof. Zengxi (Stephen) Pan | University of Wollongong | TYUT 2026 | October 2026

MATLAB files

Four starter scripts are supplied in the MATLAB folder beside this document. Use base MATLAB R2020a or later; no additional toolbox is required. Each script saves its results in MATLAB/Plots.

MATLAB/Project01_01_Input.m — Stage 1 floor input.

MATLAB/Project01_02_Response.m — Stage 4.2 floor-bump responses.

MATLAB/Project01_03_Frequency.m — Stage 5 Bode magnitude.

MATLAB/Project01_04_FreeDecay.m — Stage 4.1 fixed-floor release.